FFGFT in Literature Comparison:

What the Galois Approach Achieves and What It Does Not

Johann Pascher
ORCID 0009-0000-6518-4064

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Status markers

[K] *Confirmed* — derived completely from ξ or a corpus-secured derivation; all assertions in the verification script passed.

[B] *Established* — numerically secured by script or analytically trivial.

[S] *Speculative* — suggestion or conjecture, not yet fully proved.

Abstract

This document compares the results of the Galois documents 338–344 systematically with the established literature on algebraic derivations of the Standard Model (division algebras, Clifford algebras, exceptional Jordan algebras) and on neutrino physics (seesaw mechanism, $0\nu\beta\beta$). It names where FFGFT achieves something absent from the literature, and where the literature is ahead of FFGFT. The criterion is not “is the algebra elegant” but “does the algebra give testable numbers”.

References. All citations were verified on 4 September 2026 against arXiv, INSPIRE, APS and publisher pages.

1 The comparison class: algebraic approaches to the Standard Model

Table 1: Algebraic approaches to the Standard Model (selection)

Approach	Authors	Algebra	Core claim
Octonion quarks	Günaydin, Gürsey 1973	$\mathbb{O}$	SU(3) from octonion automorphisms
Division algebras	Dixon 1994	$\mathbb{R}\otimes\mathbb{C}\otimes\mathbb{H}\otimes\mathbb{O}$	One generation from $\mathbb{T}$
Clifford Cl(6)	Furey 2015, 2018	Cl(6) left ideals	8 fermions, SU(3)×U(1)
Exceptional Jordan	Todorov, Dubois-Violette 2018	$J_3(\mathbb{O})$	Three generations from F_4
GALG / Hyperbit	Matzke 2022–2026	Cl(n) over GF(9)	Vacuum structure, order 26
FFGFT / Galois	Pascher 2026	GF(3^k), $T^4/\mathbb{Z}_3$	Masses from ξ, primes forced

The decisive question since Günaydin 1973 has been the same: *The algebra has the right shape — does it also give the right numbers?*

2 What FFGFT achieves beyond the literature

Prime structure algebraically forced [K]

Literature. The harmonic primes $\{5,7,11,13\}$ appear in music theory (Partch, just intonation) and phenomenological mass formulae (Koide 1983, Brannen 2006) as observed, never derived. Furey and Dixon work in characteristic 0; GF(3^k) does not appear in any comparison approach. **FFGFT.** Docs. 342/343: these primes are the only ones entering $|\mathrm{GF}(3^k)^*|$ at $k = 3,4,5,6$; $p = 13$ is the worldwide unique prime with ord$_p(3) = 3$. A theorem, not a fit. **Assessment.** New in this form.

Coupling constant not a free parameter [K]

Literature. In the SM all Yukawa couplings are free (13 parameters in the fermion sector). Furey 2015 derives gauge groups, not couplings. Koide 1983 finds $Q = 2/3$ but cannot explain why. **FFGFT.** Doc. 338: $\xi = (r_\mu/r_e)^2/43200$; hence $m_\mu/m_e = 120\sqrt{3}$ (0.52 %) and $1/\alpha = 3700/27$ (7.6 ppm). **Assessment.** Concrete numerical progress beyond all comparison approaches. The torus quantum numbers $r_i = n_{\phi,g}^2/n_{\theta,g}$ and $p_g = 3/(2g)$ follow from winding numbers of the T^4 torus, which derive from GF(3^n) group orders: $n_{\theta,1} = 3$ (characteristic), $n_{\phi,1} = 2$ ($|\mathrm{GF}(3)^*|$), $n_{\theta,2} = 5$ (smallest prime factor of $|\mathrm{GF}(81)^*| = 80$) etc. (Doc. 338, Tab. 2) **[K]**. C_{conv} is the SI unit conversion, not a free parameter.

Neutrino–vacuum identity [B]

Literature. Furey 2015 constructs the neutrino as the Fock vacuum of Cl(6) — known. **FFGFT.** Doc. 344: the same follows independently from GF$(27)^*$: ν_1 in $O_1 = \{1,3,9\}$, the generator orbit. Two unrelated frameworks place the same particle at the same algebraic position. **Assessment.** The convergence is new.

Closure of the classification: no fourth generation [B]

Literature. Furey 2018 and Todorov/Dubois-Violette 2018 show "there are three", not "there cannot be four" — a construction, not a theorem. **FFGFT.** $\varphi(26) = 12 \to$ exactly 4 orbits;

sector pairing without fixed point → exactly 2 layers; orbit length 3 → exactly 3 generations. A counting argument with completeness. **Assessment.** The completeness conclusion is new in this sharpness.

Falsifiable numerical predictions [K]/[B]

Literature. Furey, Dixon, Günaydin give structural statements, no mass values. The seesaw (Minkowski 1977, Mohapatra/Senjanović 1979, Weinberg 1979) explains the smallness of neutrino mass but predicts no number — M_R is free. **FFGFT.** Doc. 340: $m_{\nu_1} = 4.54$, $m_{\nu_2} = 9.81$, $m_{\nu_3} = 49.9$ meV; Δm^2 to 0.5–1 %. Doc. 344: $m_{ee} = 0$ (Dirac) or ≈ 7.1 meV (Majorana) — nEXO decides. **Assessment.** The corpus makes concrete, falsifiable predictions no algebraic comparison approach provides. This is the main difference.

Gravity as geometry, not force [B]

Furey and Dixon do not treat gravity; their algebras yield the SM gauge groups, not spacetime. In FFGFT gravity is *not a separate force*: $G = \xi^2/(4\,m_{\text{char}})$ follows intrinsically from the $T^4/\mathbb{Z}_3$ torus geometry (Doc. 003, register R58). No separate quantum gravity is required since the geometry is pre-ordered, not a dynamical field to be quantised. The time field $T(x)$ with $T \cdot m = 1$ replaces metric coupling. The derivation runs via ξ (from Galois group orders **[K]**) and torus geometry; C_{conv} is the SI unit convention. G is therefore fully Galois-derived **[K]**.

3 What the literature achieves beyond FFGFT

Gauge groups from the algebra

Literature. Furey 2015: SU(3)×U(1) from Cl(6) commutators, correct charges for all 8 fermions. Dixon 1994: the full SU(3)×SU(2)×U(1) from $\mathbb{T} = \mathbb{R} \otimes \mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$.

FFGFT. Doc. 339 **[B]**: SU(3) from $4\,\mathbb{Z}_{13}$-orbits $\times\, 2\,\mathbb{Z}_2$-parities $= 8$ gluons; U(1) from fixed field GF(3) (photon). Doc. 336 **[K]**: Gal$(GF(9)/GF(3)) = \mathbb{Z}_2$ explains algebraically why SU$(2)_L$ and U$(1)_Y$ exist; $\sin^2\theta_W \approx 0.250$ tree-level **[K]**. Doc. 346 **[B]**: $Q \in \{0, \pm\frac{1}{3}, \pm\frac{2}{3}, \pm 1\}$ from Legendre symbol mod 13. Doc. 347 **[B]**: $Q = I_3 + Y/2$ from jE7 projector and Legendre symbol. Doc. 349 **[B]**: SU$(2)_L$ doublet structure from Cl(6) grade parity — Su (even grades) $=$ right-handed, Sd (odd grades) $=$ left-handed; $\mathbb{Z}_3$ symmetry of the torus forces the chirality split.

Balance. Charge content, Gell-Mann–Nishijima, and chirality structure fully derived from Galois/Clifford algebra **[B]**. What Furey/Dixon have more completely: SU(3)×U(1) as Clifford commutator and the octonion colour construction — a complementary framework, no contradiction. **At gauge-group content now broadly equal.**

Quark sector

Literature. Furey/Dixon treat quarks on equal footing; colour charge as octonion automorphism with full SU$(3)_c$ representation.

FFGFT. Quark masses in Doc. 006 **[K]** (deviations 0.0–0.3 % for up, down, strange, charm, bottom). Doc. 346 **[B]**: N mod 3 colour charge classifies quarks (triplet) vs. leptons (singlet). Charge values $Q = \pm\frac{2}{3}, \pm\frac{1}{3}$ from Doc. 347 **[B]**. Doc. 348 **[B]**: coupling matrix $f_3 \times f_4$ (leptons × quarks) $=$ permutation matrix, Gen.2 ↔Gen.3 forced; CKM mixing angles **[S]** (Doc. 348).

Balance. Colour charge classification and quark masses from Galois **[B][K]**. What Furey/Dixon have more completely: octonion SU$(3)_c$ construction and full CKM motivation. **At quark charges equal; at gauge construction literature ahead.**

Seesaw mechanism. Explains *why* neutrinos are light and provides a dynamical mechanism for Majorana masses. FFGFT replaces this by double-ξ suppression (Doc. 007) — shorter, but without dynamics. **Literature mechanistically ahead.**

Rigour. Furey 2015 is a thesis with complete proofs. The FFGFT corpus uses verification scripts and numerical checks (**[B]**), not complete analytic proofs for all theorems. **Literature formally stricter.**

4 The Yukawa Lagrangian and the Flavour Problem

In the Standard Model fermion masses arise from the Yukawa term:

$$\mathscr{L}_{\text{Yukawa}} \supset -y_{ij}\bar{\psi}_i H\psi_j + \text{h.c.}, \quad m_i = y_i \times \frac{v}{\sqrt{2}}.$$

The Yukawa couplings y_i are **free parameters** — measured, not derived. This is the flavour problem: 13 free fermion masses without algebraic origin.

Furey and Dixon derive the gauge structure but say nothing about the Yukawa couplings. The Lagrangian retains free y_i.

Froggatt-Nielsen mechanism (1979) [24]. A U(1) flavour symmetry assigns charge q_i to each field; couplings scale as $y_{ij} \sim \varepsilon^{|q_i+q_j|}$. This explains the hierarchy — but the charge assignments q_i are chosen freely.

FFGFT mass formula. FFGFT replaces the Lagrangian approach by a direct geometric derivation:

$$m_i = r_i\,\xi^{p_i}\,v,$$

where $r_i = n_{\phi,g}^2/n_{\theta,g}$ and $p_g = 3/(2g)$ follow from torus winding numbers forced by GF(3^n) group orders **[K]** (Doc. 338), and $\xi = 4/30000$ is algebraically derived **[K]**.

Table 2: Froggatt-Nielsen vs. FFGFT

Criterion	Froggatt-Nielsen	FFGFT
Hierarchy principle	ε^{q_i}, q_i free	ξ^{p_i}, p_i forced **[K]**
Small parameter	ε: free	$\xi = 4/30000$: Galois-derived **[K]**
Flavour symmetry	U(1), freely chosen	GF(3^k) group structure **[K]**
Mass values	order of ε, not exact	0.0–0.5 % agreement **[K]**
Three generations	unexplained	orbit length 3 forced **[B]**

FFGFT and Froggatt-Nielsen share the same core principle (power law of a small parameter), but FFGFT forces everything algebraically where FN freely assigns. This is the precise advance over the closest comparison in the literature.

5 The central difference

Furey, Dixon, Günaydin show: *the algebra has the right shape.*
FFGFT shows: *the algebra gives the right numbers* — $m_{\nu_1} = 4.54$ meV, Δm^2 to 1 %, $1/\alpha$ to 7.6 ppm, m_μ/m_e to 0.5 %.

Not a contradiction but a complement. Furey provides the gauge structure, the FFGFT Galois structure provides the mass scale. Doc. 344 shows both converge at Su[0] $=$ Vss[0]. A unification — Cl(6) over GF(9) with GF$(27)^*$ orbit structure — is the natural next task.

6 Honest assessment

Table 3: Comparison in one line each

Criterion	Literature	FFGFT
Gauge groups	complete (Furey, Dixon)	charges+chirality from Doc. 346/347/349 **[B]**; Clifford constr. Furey
Mass values	none	concrete, 0.5–1 %
Neutrino masses	mechanism (seesaw), no value	values, no mechanism
Prime structure	observed	derived **[K]**
Generation count	constructed	counted (complete) **[B]**
Falsifiability	weak (structural)	sharp (nEXO, Δm^2)
Quarks	equal footing	secondary
Proof form	analytic (structure)	numerical (numbers) — different domains
Gravity	absent	$G = \xi^2/(4m_{\text{char}})$ from torus geometry (R58)

FFGFT complements Furey/Dixon with the mass scale. The gauge structure is more completely worked out in the literature; concrete, falsifiable mass values to 0.5–1 % are not present in any comparison approach.

Note on AI assistance

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